

Predictive Modelling

Philip Hans Franses
EQI, 2025

5: Trends

Outline

Trends in AR models

Deterministic trend

Stochastic trend

Testing for a unit root

A trend

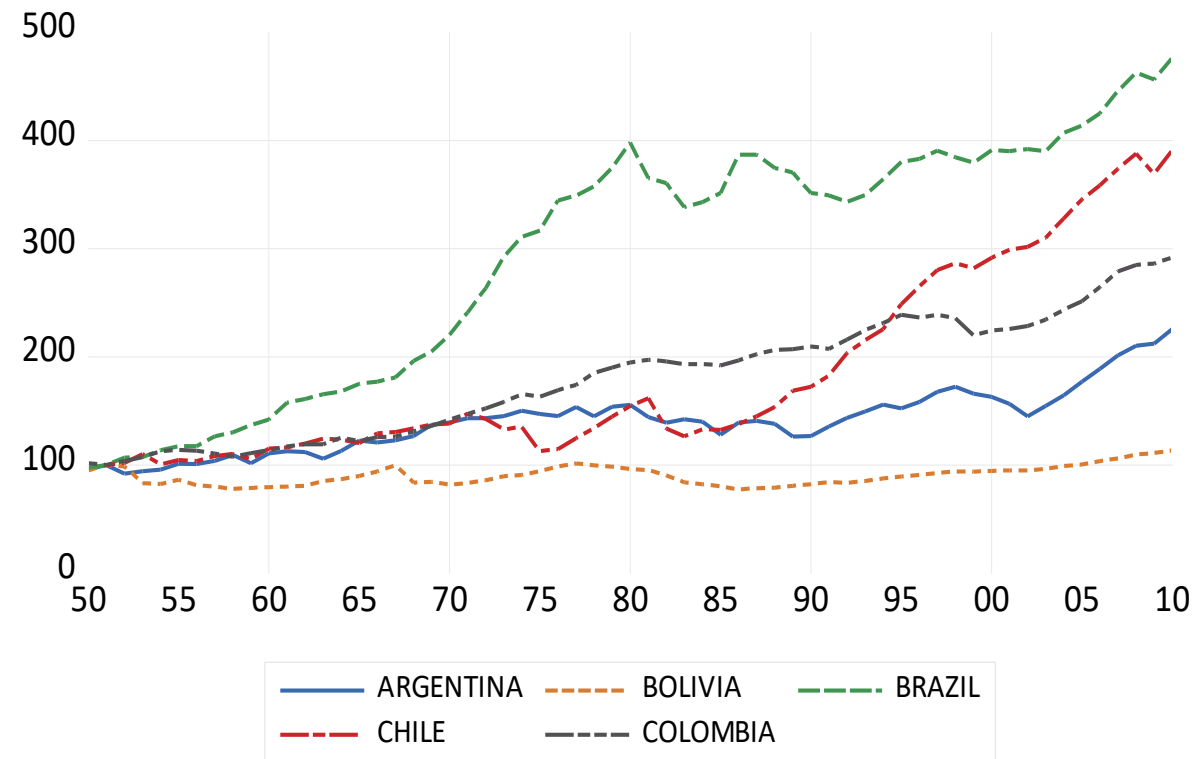
A trend is a general tendency for a variable to decrease or increase over time

Why is it important to study trend properties?

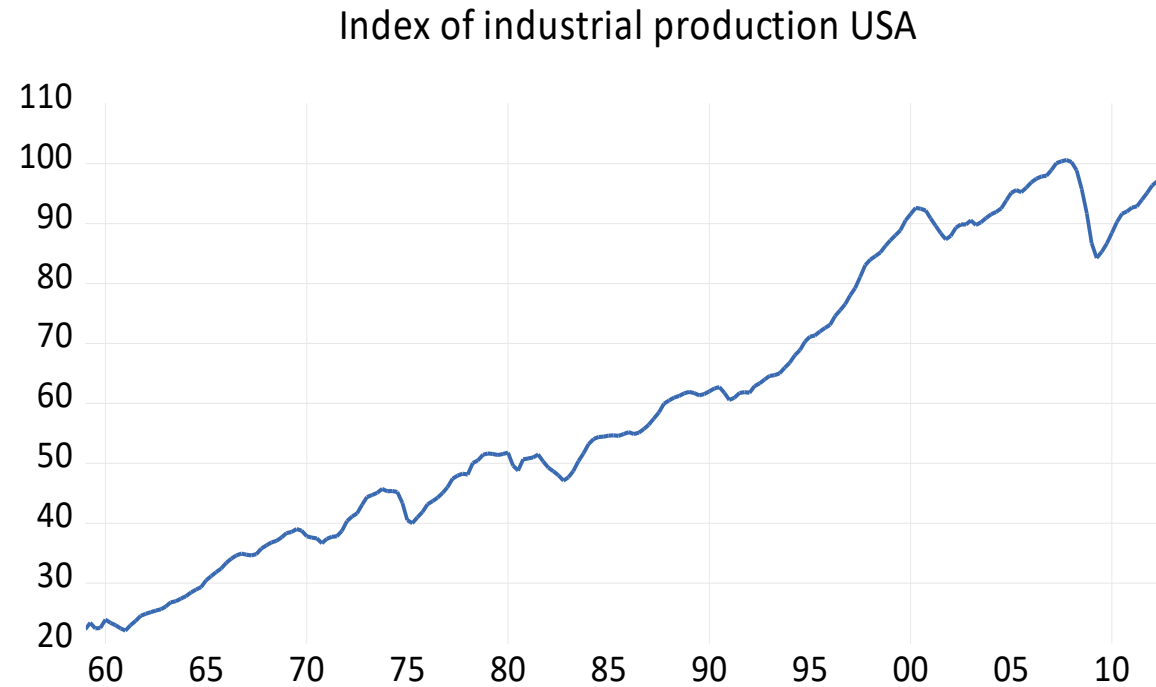
A trend is a dominant feature of time series.

For accurate (long-term) forecasts, it is crucial to describe the trend properties of a time series correctly.

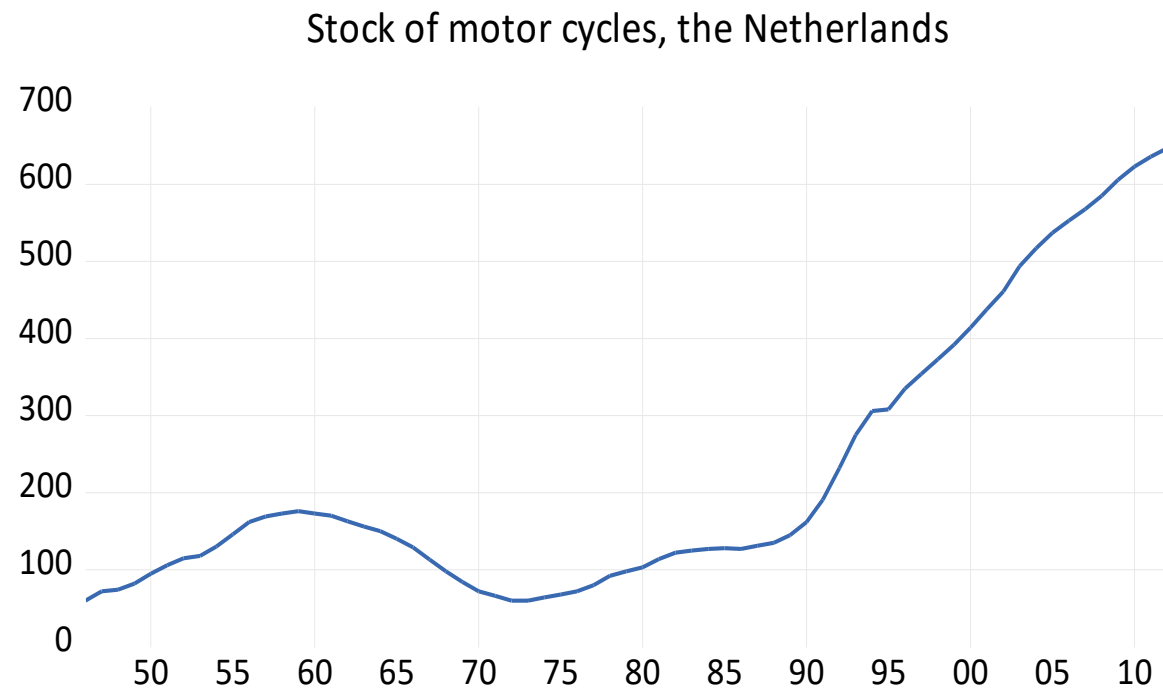
Look again at



Trend, expansions and recessions



Trend with different slopes



Trend in AR models

Consider the AR(1) model

$$y_t - \mu = \phi_1(y_{t-1} - \mu) + \varepsilon_t$$

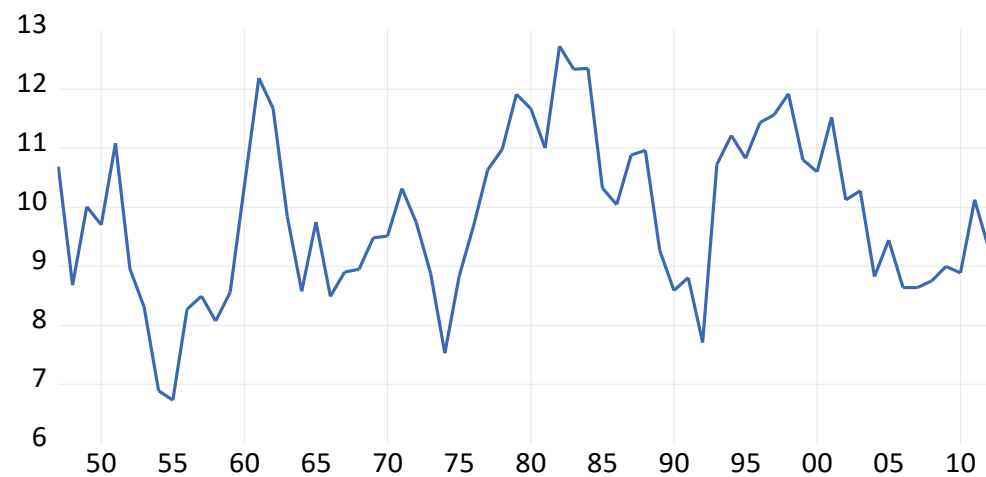
$$t = 1, 2, \dots, T.$$

When $|\phi_1| < 1$, y_t has an unconditional mean which is equal to μ for all t , has an unconditional variance which is constant and equal to $\frac{\sigma^2}{1-\phi_1^2}$, and it has constant autocorrelations.

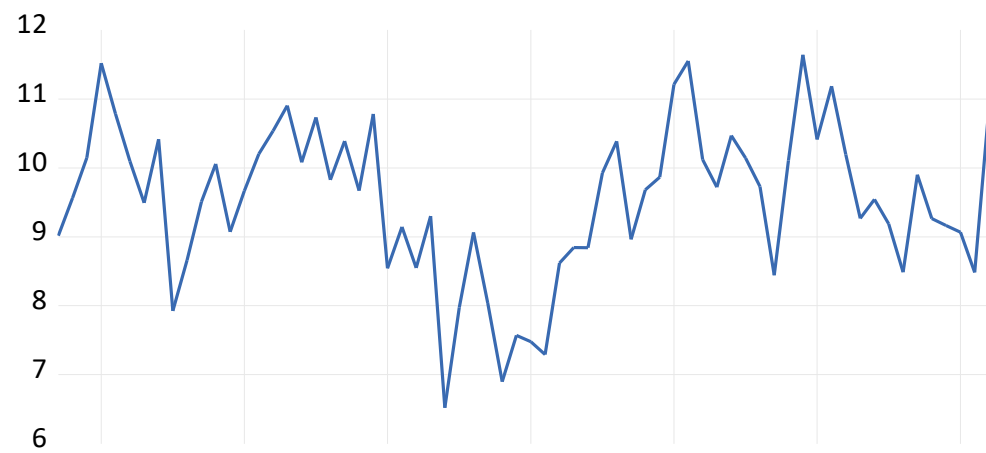
These three properties form the definition of stationarity. This AR(1) model cannot describe data with a trend (for which the unconditional mean changes with every new observation)

No trend in a stationary AR(1): $\phi_1 = 0.8, \mu = 10$

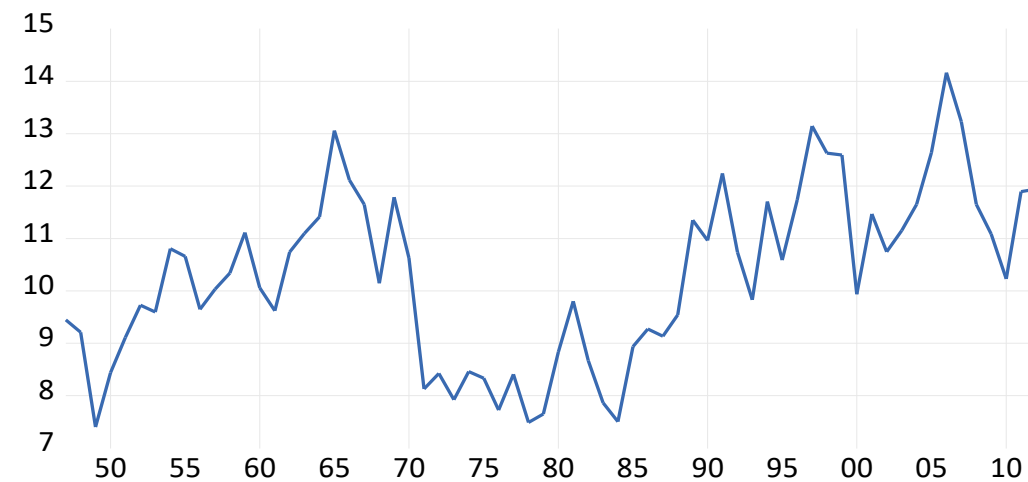
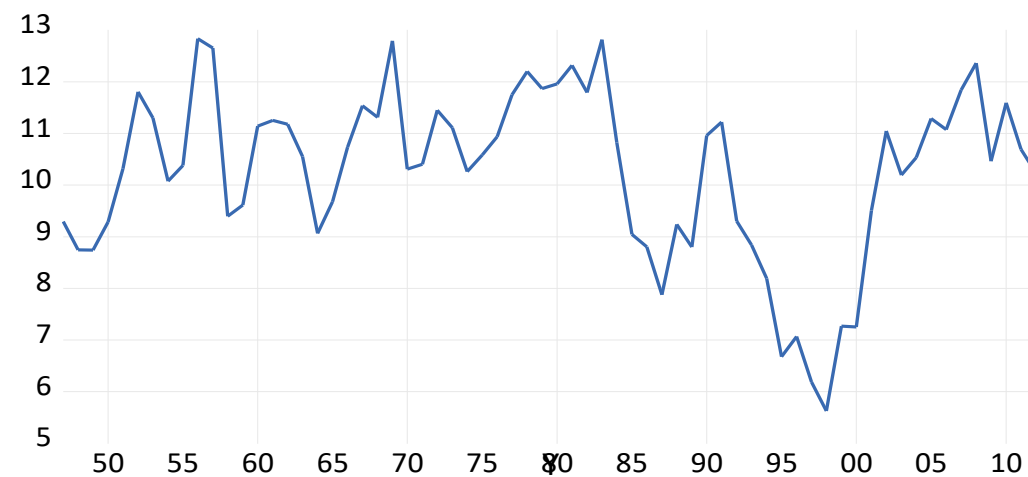
Y



Y



Y



Trends in AR models

Two possibilities to incorporate trends in AR models:

Insert a deterministic trend (next to the constant)

Set $\phi_1 = 1$ to have a stochastic trend

Deterministic trend in AR models

Consider again the AR(1) model, but now with an additional linear trend term:

$$y_t - \mu - \delta t = \phi_1(y_{t-1} - \mu - \delta(t-1)) + \varepsilon_t$$

and assume that $|\phi_1| < 1$.

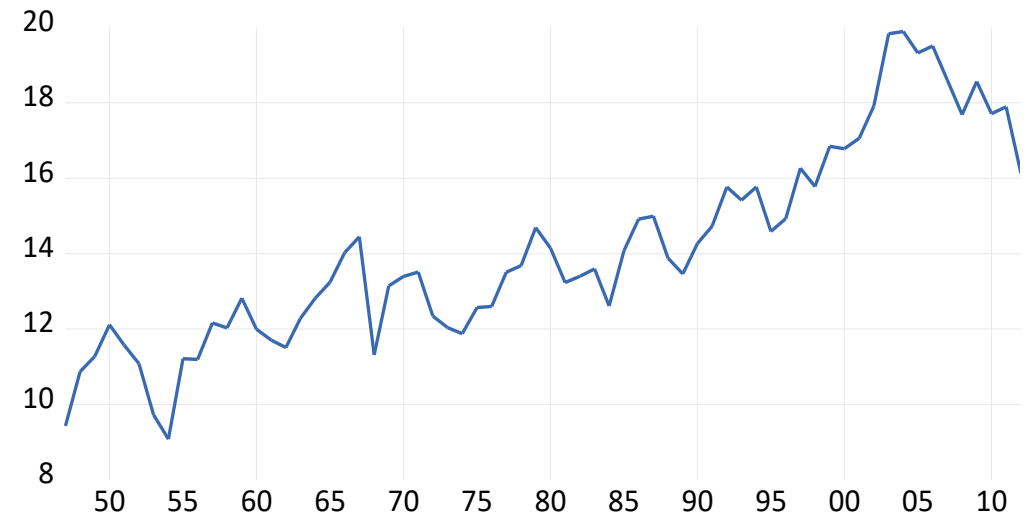
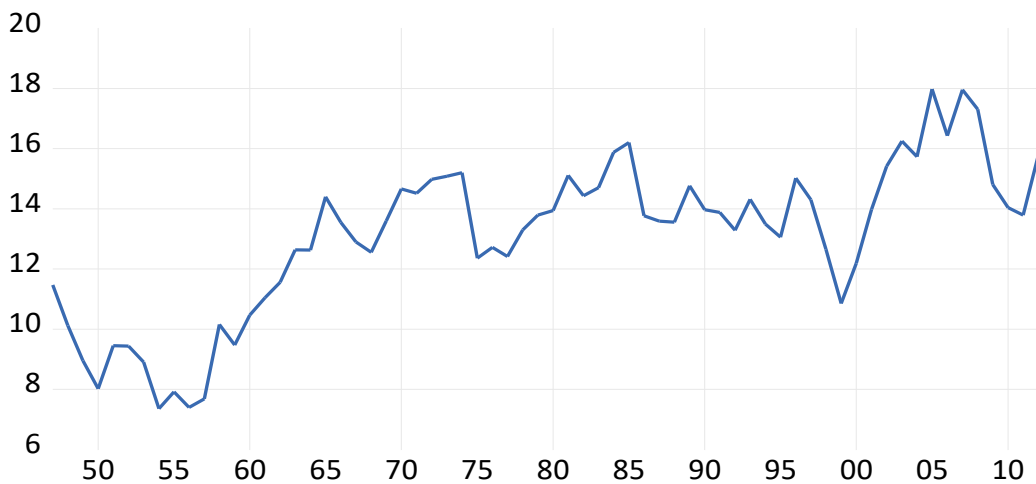
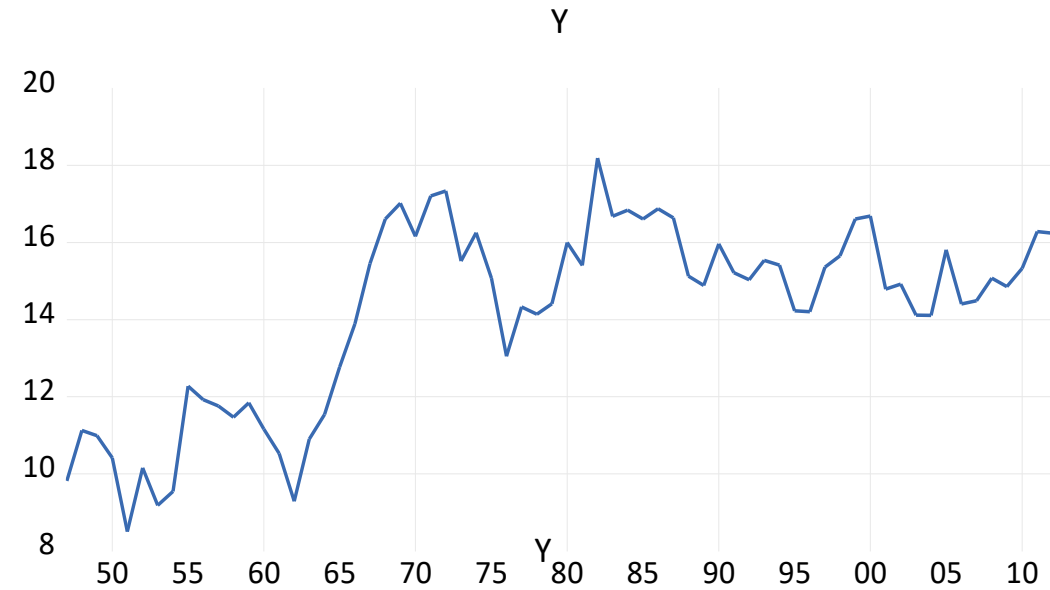
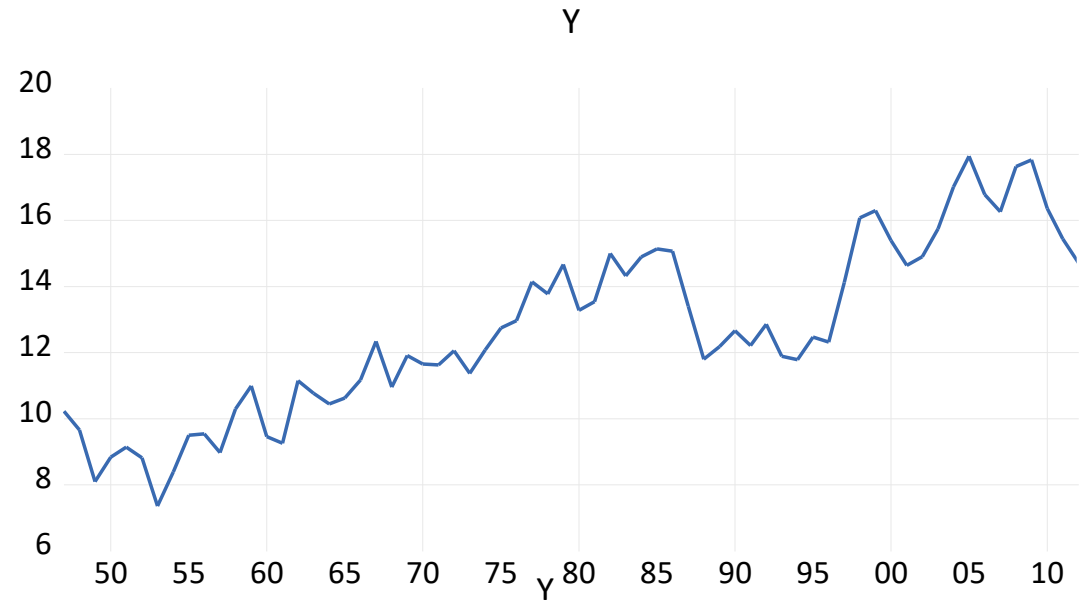
Define $z_t = y_t - \mu - \delta t$, then we have

$$z_t = \phi_1 z_{t-1} + \varepsilon_t$$

with an unconditional mean which is equal to 0 for all t , with an unconditional variance which is constant and equal to $\frac{\sigma^2}{1-\phi_1^2}$, and it has constant autocorrelations. From $E(z_t) = 0$ follows that

$$E(y_t) = \mu + \delta t$$

Trend in a stationary AR(1): $\phi_1 = 0.8, \mu = 10, \delta = 0.1$



Trend-stationarity

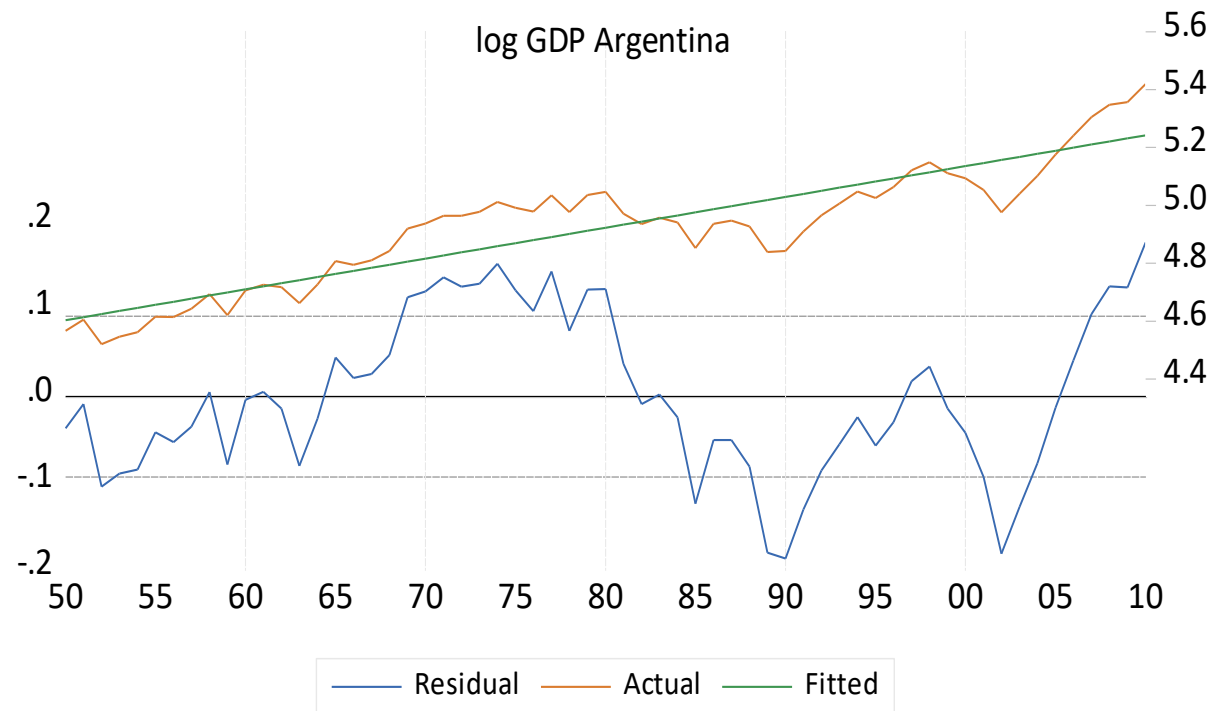
Because of

$$z_t = \phi_1 z_{t-1} + \varepsilon_t$$

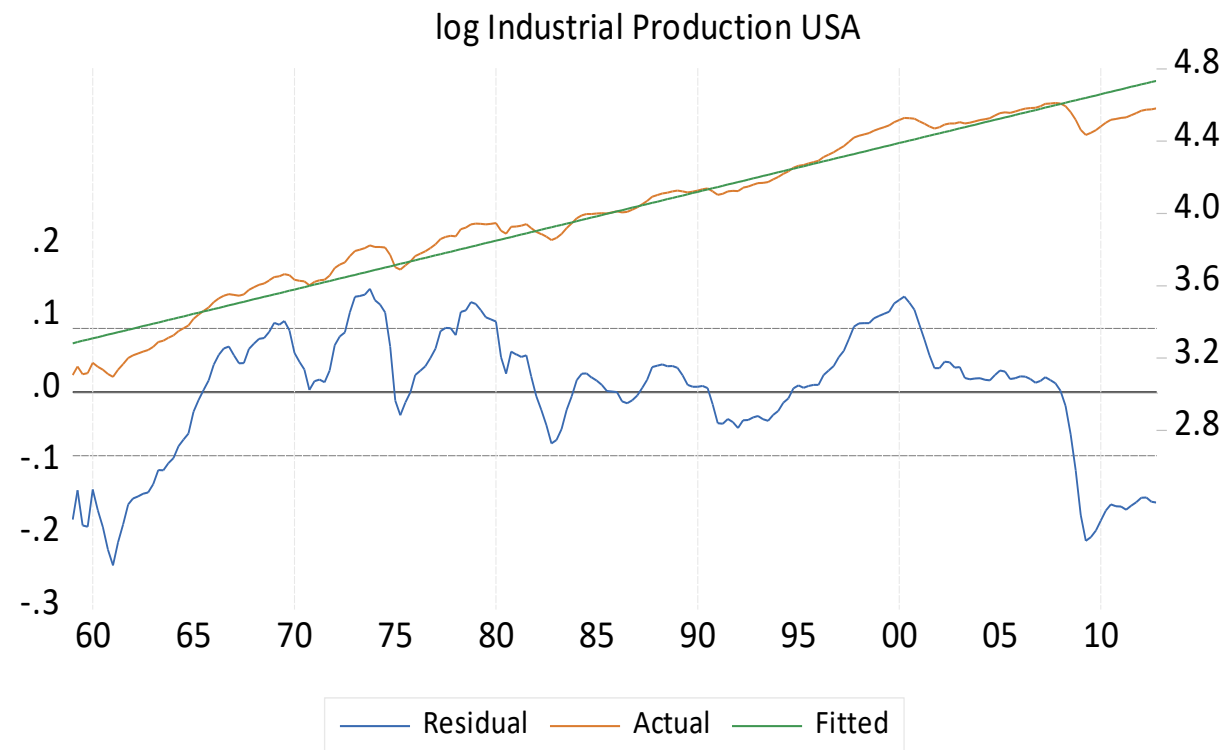
with $|\phi_1| < 1$, z_t mean-reverts to the unconditional mean 0. Hence, y_t mean-reverts to (“is attracted by”)

$$\mu + \delta t$$

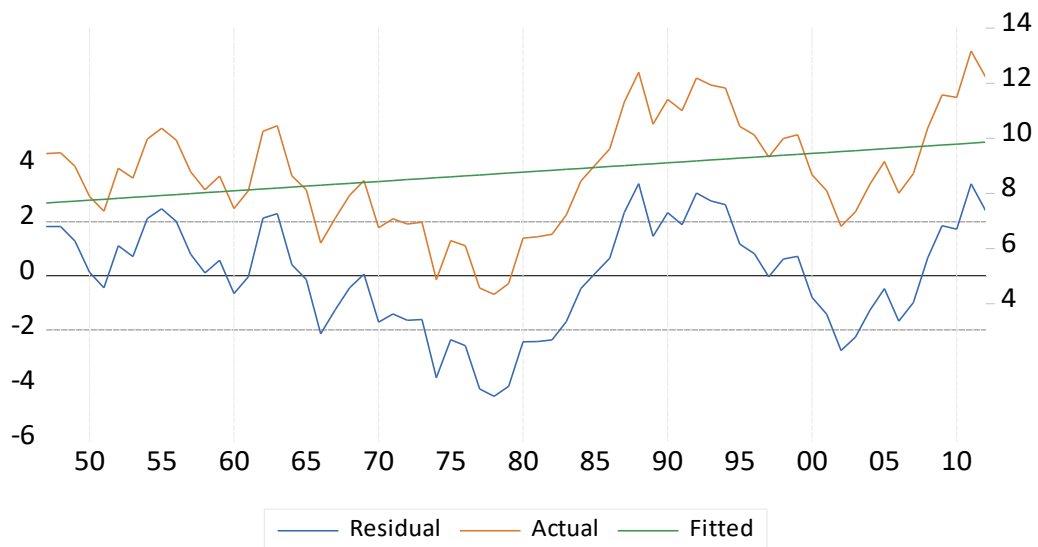
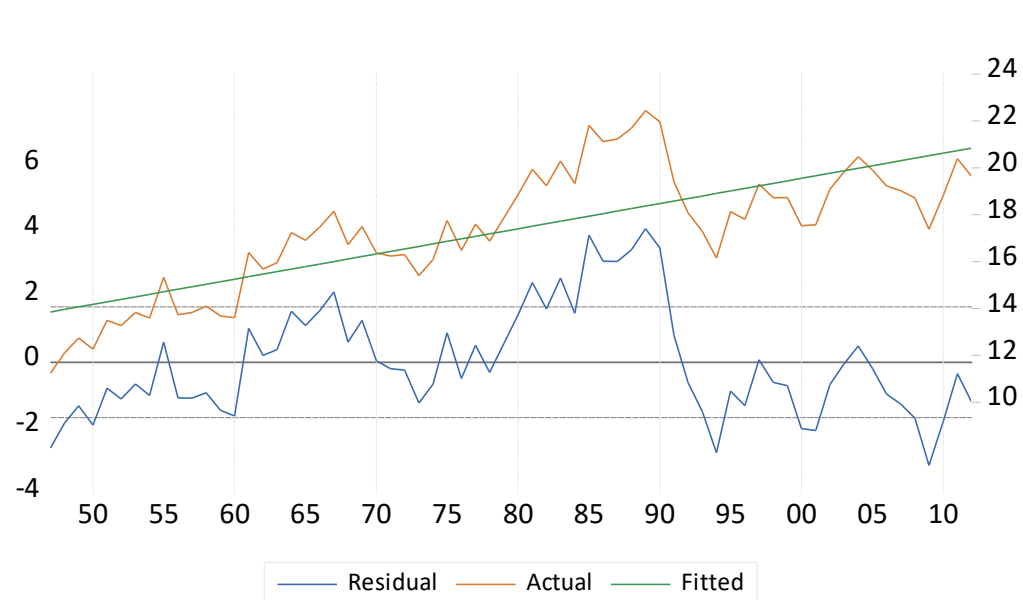
Trend-stationarity?



Reversion to trend?



Trend in a stationary AR(1): $\phi_1 = 0.99, \mu = 10, \delta = 0.1$



Stochastic trend in AR models

Consider again

$$y_t - \mu - \delta t = \phi_1(y_{t-1} - \mu - \delta(t-1)) + \varepsilon_t$$

and now set $\phi_1 = 1$. Then

$$y_t - \mu - \delta t = y_{t-1} - \mu - \delta(t-1) + \varepsilon_t$$

$$y_t - \delta t = y_{t-1} - \delta t + \delta + \varepsilon_t$$

$$y_t = y_{t-1} + \delta + \varepsilon_t$$

Did the trend disappear?

No, because

$$\begin{aligned}y_t &= y_{t-1} + \delta + \varepsilon_t \\y_t &= (y_{t-2} + \delta + \varepsilon_{t-1}) + \delta + \varepsilon_t \\y_t &= (y_{t-3} + \delta + \varepsilon_{t-2}) + 2\delta + \varepsilon_{t-1} + \varepsilon_t \\&\vdots \\y_t &= y_0 + \delta t + \sum_{i=1}^t \varepsilon_i\end{aligned}$$

This is called a stochastic trend model

Mean and variance

$$y_t = y_0 + \delta t + \sum_{i=1}^t \varepsilon_i$$

The unconditional mean is

$$E(y_t) = y_0 + \delta t$$

The unconditional variance is

$$V(y_t - y_0 - \delta t) = V\left(\sum_{i=1}^t \varepsilon_i\right) = E\left(\sum_{i=1}^t \varepsilon_i^2\right) = t\sigma^2 = \gamma_{0,t}$$

What does that mean for forecasting?

Forecast intervals become very wide.

Recall that for an AR(1):

$$V(e_{T+h|T}) = (1 + \phi_1^2 + \phi_1^4 + \dots + \phi_1^{2(h-1)})\sigma^2$$

When $|\phi_1| < 1$, then $\phi_1^{2(h-1)} \rightarrow 0$. But when $\phi_1 = 1$, then

$$V(e_{T+h|T}) = h\sigma^2$$

Autocorrelations

Likewise,

$$\begin{aligned}\gamma_{1,t} &= E(y_t - y_0 - \delta t)(y_{t-1} - y_0 - \delta(t-1)) \\ &= E\left(\sum_{i=1}^t \varepsilon_i \sum_{i=1}^{t-1} \varepsilon_i\right) = (t-1)\sigma^2\end{aligned}$$

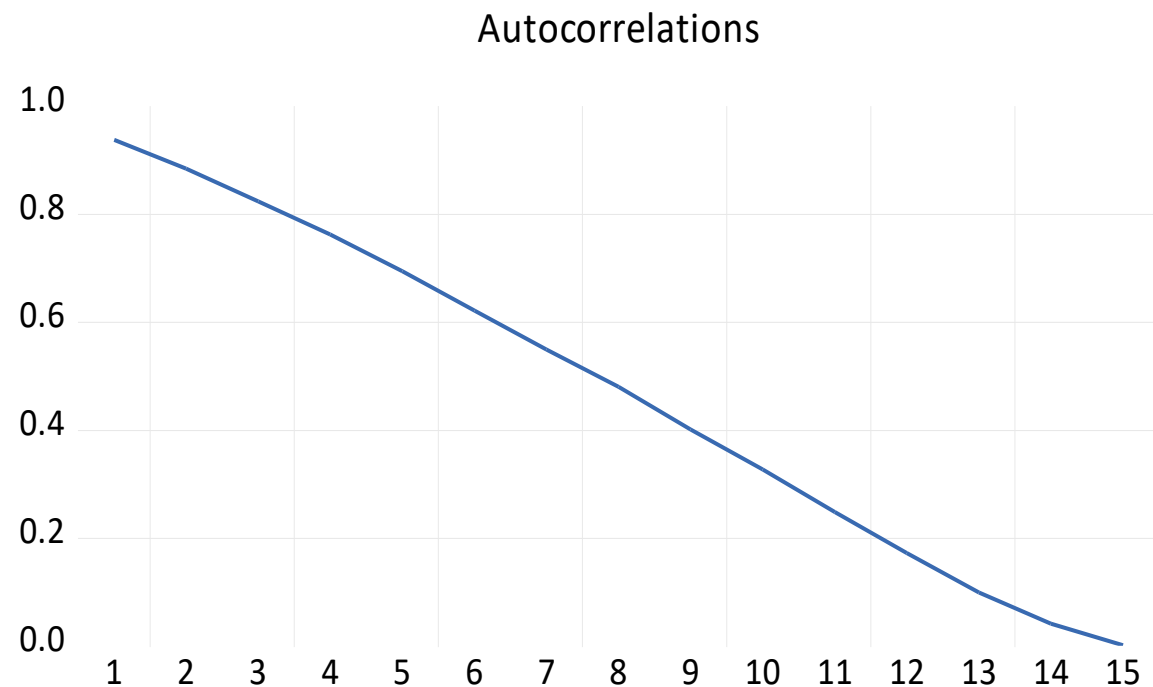
Hence,

$$\rho_1 = \frac{t-1}{t}$$

And also

$$\rho_k = \frac{t-k}{t}$$

This explains for a non-stationary AR(1): $\phi_1 = 1, \delta = 0.1$



Difference-stationarity

If

$$y_t = y_{t-1} + \delta + \varepsilon_t$$

then

$$y_t - y_{t-1} = \delta + \varepsilon_t$$

Thus: go from levels to growth rates (differences) when $\phi_1 = 1$

Higher-order AR?

$$y_t - y_{t-1} = \delta + \varepsilon_t$$

becomes for an AR(2) for y_t

$$y_t - y_{t-1} = \pi_1(y_{t-1} - y_{t-2}) + \delta + \varepsilon_t$$

Rearranging gives

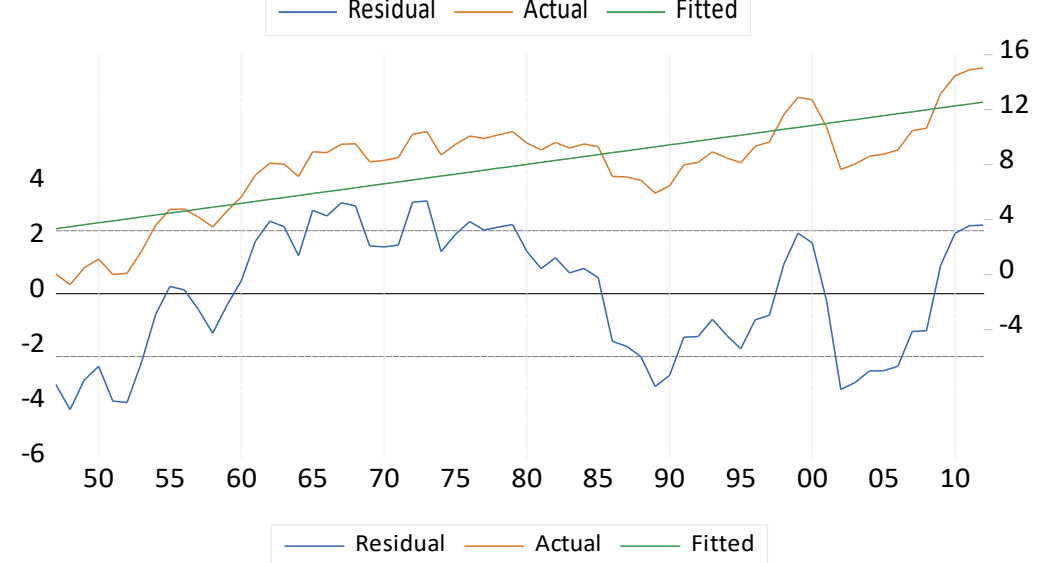
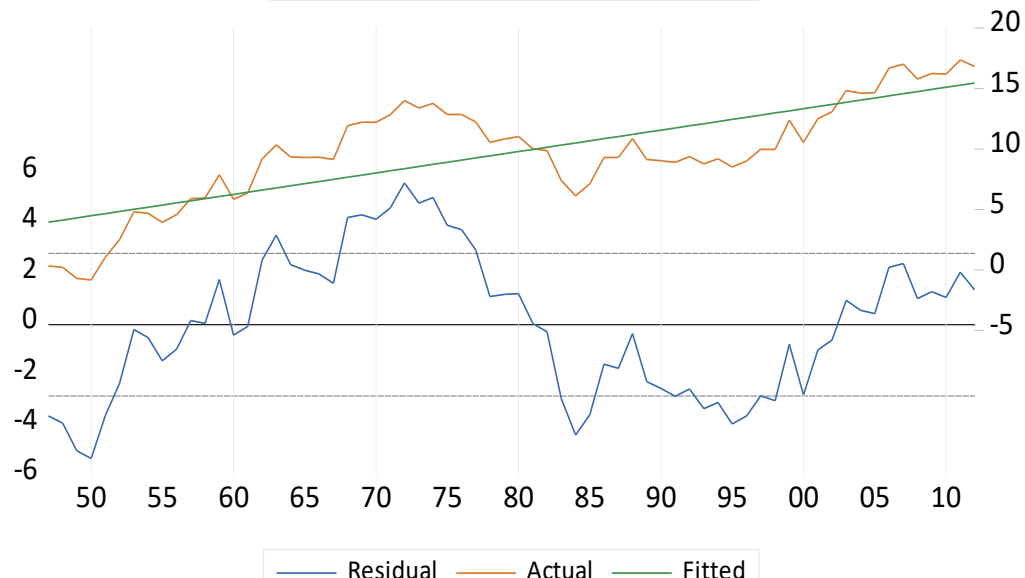
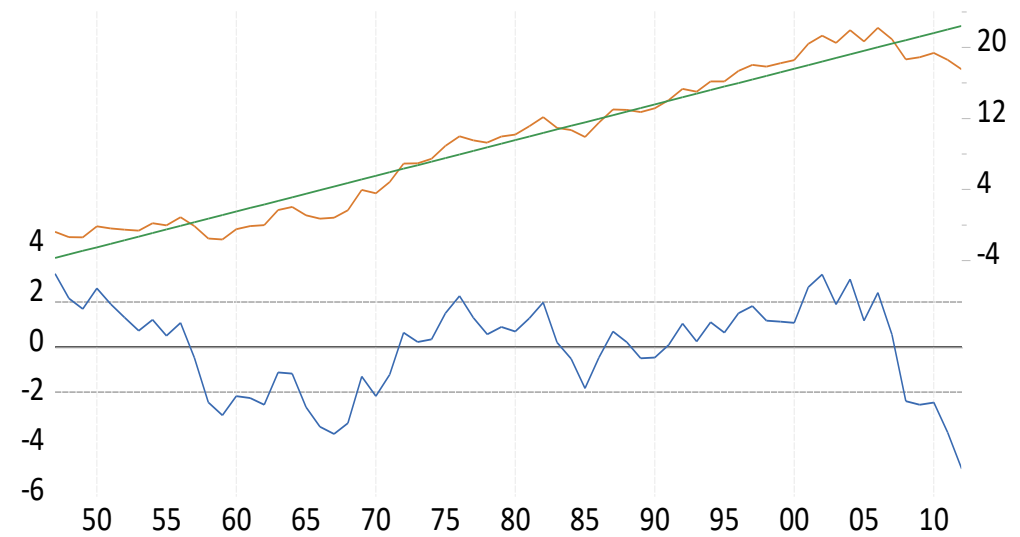
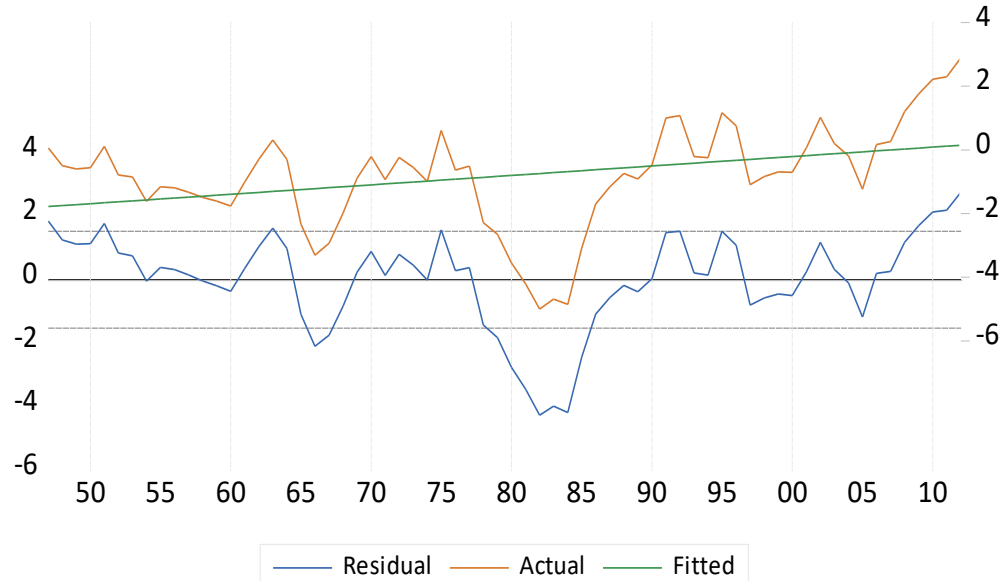
$$y_t = (\pi_1 + 1)y_{t-1} - \pi_1 y_{t-2} + \delta + \varepsilon_t$$

This shows that a stochastic trend in an AR(2) for y_t like

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$$

means that the **sum** of the AR parameters equals 1, that is, $\phi_1 + \phi_2 = 1$.

Stochastic trend: AR(1) with $\phi_1 = 1, \delta = 0.1$



Exercise in R

Motivation: See that series x and y generated from DT and ST models can have very similar patterns.

Also understand role of magnitude of AR-parameter.

Repeat with ϕ_1 equal to 0.7, 0.9 and 0.95 in the model for x_t .

Distinguishing between DT and ST models

Important differences between deterministic trend [DT] and stochastic trend [ST] models are:

The impact of shocks is transitory for the DT model and permanent for the ST model.

The out-of-sample forecast error variances for the ST model are much larger than those for the DT model. Hence, ST data may be more difficult to forecast.

The statistical properties of ST data differ from those of stationary time series. This means that alternative asymptotic theory is needed.

Main idea

Consider

$$y_t = \phi_1 y_{t-1} + \varepsilon_t$$

and test

$$H_0: \phi_1 = 1$$

versus (one-sided)

$$H_1: \phi_1 < 1$$

This is called a test for a unit root.

Main idea: to make it a standard t test

Consider

$$\begin{aligned}y_t - y_{t-1} &= (\phi_1 - 1)y_{t-1} + \varepsilon_t \\ &= \rho y_{t-1} + \varepsilon_t\end{aligned}$$

and test

$$H_0: \rho = 0$$

Versus (one-sided)

$$H_1: \rho < 0$$

Non-standard distribution of test

The t test statistic on $\rho = 0$ does not follow an asymptotic standard normal distribution.

Its distribution is shifted to the left.

Critical values have to be simulated.

Tables are available for different sample sizes and different models.

Critical values are the same when additional lagged differences are added.

Critical values (sample is ∞)

Model (possibly with added lags of $\Delta_1 y_t$)	1%	5%	10%
$\Delta_1 y_t = \rho y_{t-1} + \varepsilon_t$	-2.58	-1.95	-1.62
$\Delta_1 y_t = \rho y_{t-1} + \alpha + \varepsilon_t$	-3.43	-2.86	-2.57
$\Delta_1 y_t = \rho y_{t-1} + \alpha + \beta t + \varepsilon_t$	-3.96	-3.41	-3.12

Rejection of $H_0: \rho = 0$ if t test statistic is to the **left** of the critical values
(read: more negative)

Logs of USA industrial production (seasonally adjusted)

$$\Delta_1 y_t = \rho y_{t-1} + \alpha + \beta t + \pi_1 \Delta_1 y_{t-1} + \pi_2 \Delta_1 y_{t-2} + \varepsilon_t$$

t test statistic for $H_0: \rho = 0$ has value -0.434. This is to the right of the 5% critical value -3.41, and hence $H_0: \rho = 0$ cannot be rejected.

Continue with growth rates instead of levels.

Returns on the Dow Jones index

$$\Delta_1 y_t = \rho y_{t-1} + \alpha + \varepsilon_t$$

t test statistic for $H_0: \rho = 0$ has value -64.447. This is to the left of the 5% critical value -2.86, and hence $H_0: \rho = 0$ is rejected.

Exercise in R: What happens if the data have a structural break?

Motivation: See that the level shift that occurs in x has an effect that depends on the value of an $AR(1)$ parameter, even such that we may think that this series has a unit root.

Do a unit root test for y and for x . What happens if add becomes 15 for observations 150 to 300?

What did we learn?

Trends in AR models: depends on the AR parameters

Compared with a deterministic trend model, the stochastic trend model has much wider confidence intervals

Testing for a unit root requires new asymptotic theory.

In practice: in case of doubt whether to start with levels or growth rates (differences), do a unit root test.